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SUPERSEDING
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FEDERAL SPECIFICATION

TAPE, EMBOSSING

This Federal Specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for metal and plastic tapes for use in embossing machines to produce raised letters, figures, and symbols.

1.2 Classification.

1.2.1 Types and classes. The embossing tape shall be of the following types and classes as specified (see 6.2):

Type I - Plastic, with adhesive and liner.
Class 1 - Clear.
Class 2 - Colored.
Class 3 - Fluorescent.

Type II - Metal, with adhesive and liner.
Class 1 - Aluminum.
Class 2 - Copper.
Class 3 - Deleted (see 6.3).

Type III - Metal, without adhesive or liner.
Class 1 - Aluminum.
Class 2 - Copper.
Class 3 - Deleted (see 6.3).
Class 4 - Stainless steel.
Class 5 - Monel.
Class 6 - Zinc.

1.2.2 Color. (See 3.3 and 6.2.)

1.2.3 Finish. (See 3.4 and 6.2.)

1.2.4 Width and length. (See 3.6, 3.7, and 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

OO-E-491 - Embossing Machines, Identification Tape, (Manually Operated).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal, Edged and Components.
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Test Method Std. 101/GEN (General Provisions) - Preservation, Packaging and Packing Materials: Test Procedures.
Fed. Test Method 101/2050 - Adhesion of Pressure-Sensitive Tape.
Fed. Test Method 101/2054 - Holding Power of Pressure-Sensitive Tapes.
Fed. Test Method 101/4050 - Weatherometer Aging of Pressure-Sensitive Tapes.
Fed. Test Method 101/4047 - Accelerated Aging of Pressure-Sensitive Packaging Tapes.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).
Fed. Test Method Std. 147/GEN (General Provisions) - Tapes, Pressure-Sensitive and Gummed; Methods of.
Fed. Test Method 147/4 - Preparation for Testing.
Fed. Test Method 147/36 - Thickness.
Fed. Std. 595 - Colors.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-3891 - Luminescent Materials and Equipment (Nonradioactive).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications: The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Backing.

3.1.1.1 Plastic (type I). The backing of the type I tape shall be a plastic film formulated to meet the requirements specified herein.

3.1.1.2 Metal (types II and III). The backing of the types II and III tapes shall be metal of the supplier's standard commercial material for the type and class specified (see 6.2).

3.1.2 Adhesive (types I and II). One side of types I and II tape shall be completely and evenly coated with a pressure-sensitive adhesive which requires no moisture, heat, other special preparation prior to application of the tape to a surface.

3.1.3 Liner (types I and II). The adhesive on types I and II tape shall be protected by a smooth liner. The liner shall be of such material to permit removal of the liner without breaking or causing separation of the adhesive from the backing.

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3.2 Transparency (type I, class 1). The type I, class 1 tape shall be sufficiently clear to allow easy reading of 10 point (1/8 inch) type when tested in accordance with 4.4.2.5.

3.3 Color. The color of the type I, class 2 tape shall be black, yellow, orange, purple, blue, green, red, brown, gray, or gold, as specified in the contract or order (see 6.2) and shall approximate the corresponding color card number of Fed. Std. No. 595 as listed in table I. The type I, class 3 flourescent tape shall be blue, green, red, orange or yellow, as specified (see 6.2).

3.3.1 Luminance. The type 1, class 3 flourescent tape shall pass the luminance requirement for type F, form 5 material in MIL-L-3891 when tested as specified in 4.4.2.6.

3.4 Finish. The type I tape shall have either a glossy finish or semi-gloss (matte) finish as specified (see 6.2).

TABLE I. Color

Color	Fed. Std. No. 595 number designation	
	Glossy finish	Semi-gloss (matte finish)
Black	17038	27038
Yellow	13538	23655
Orange	12246	22246
Purple	17142	27142
Blue	15123	25102
Green	14110	14110 1/
Red	11105	21105
Brown	10075	20059
Gray	16251	26280
Gold	17043	17043 1/

1/ Except that this numeric designation shall be in semi-gloss (matte finish) as determined by visual examination.

3.5 Physical properties. The embossing tape shall conform to the requirements specified in table II when tested as specified.

TABLE II. Physical requirements

Characteristic	Requirements		Test Paragraph
	Minimum	Maximum	
Thickness of tape including liner (mils): (types I and II only)	12.0	18.0	4.4.2.1
Thickness of tape without liner (mils): 1/			
Type I, all classes	7.5		4.4.2.1
Types II and III, class 2 (copper)	7.2		4.4.2.1
Types II and III, class 1 (aluminum)	5.4		4.4.2.1
Types III, classes 4 and 5 (stainless steel and monel)	4.5		4.4.2.1
Type III, class 6 (zinc)	9.0		4.4.2.1
Adhesion, oz. per inch width			
Types I and II only			
Initial to steel	80		4.4.2.2
Holding power, creeping or slippage, inch: (types I and II only)		1/8	4.4.2.3
Embossing	Pass 2/		4.4.2.4

1/ The thickness of the plastic tape (type I) shall be measured with adhesive. The thickness of all metal tape shall be measured without adhesive.

2/ The tape shall produce raised, well defined characters. Characters produced on plastic tape shall be white, unless otherwise specified (see 6.2).

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TABLE II. Physical requirements (con.)

Characteristic	Requirements Minimum	Maximum	Test Paragraph
Resistance to accelerated weathering:			
Colorfastness (type I, all classes)	Pass 3/		4.4.2.7
Transparency	Pass		4.4.2.7
Legibility	Pass 4/		4.4.2.7
Resistance to accelerated aging:			
Adhesion to steel, oz. per inch width (types I and II only)	64.0		4.4.2.8
Adhesive flow (types I and II only)	Pass 5/		4.4.2.8
Colorfastness (type I only)	Pass 3/		4.4.2.8
Liner removal (types I and II only)	Pase 6/		4.4.2.8

3/ The plastic tape shall show "fair" colorfastness or better when tested as specified in 4.4.2.7.
4/ The characters shall be clearly legible at a distance of 3 feet and there shall be no evidence of cracking in the plastic tape. Characters on plastic tape shall retain contrast from original color.
5/ There shall be no excessive tackiness or oozing of adhesive from the roll when tested and examined in accordance with 4.4.2.8.
6/ The liner shall not be difficult to remove. The liner shall not break or tear, nor shall it, upon removal, cause any adhesive to delaminate from the backing when tested as specified in 4.4.2.8.

3.6 Width. The width of the tape shall be as specified (see 6.2). The tolerance shall be minus 0.005 inch and plus 0.002 inch for widths of 1/2 inch or less, and ± 0.010 inch for widths greater than 1/2 inch.

3.7 Length. The minimum length of the tape shall be 144 inches, unless otherwise specified (see 6.2). The tape shall be one continuous piece without splices.

3.8 Form. Unless otherwise specified, tape shall be furnished in rolls (see 6.2). When tape is to be furnished in cartridges or reels, the tape and cartridges or reels shall conform to the requirements of 3.8.2 or 3.8.3.

3.8.1 Rolls. The tape shall be evenly and uniformly wound on itself with the liner side in to form rolls not exceeding 2 inches in diameter. The outside end of the embossing tape shall be secured with pressure sensitive adhesive tape to restrain the roll until released. The securing tape shall not extend beyond the edge of the roll and shall be easily removable without breaking or tearing.

3.8.2 Cartridges. When furnished in cartridges, the tape shall be coiled within the cartridge so that the tape is easily and properly dispensed without binding or sticking when used in the embossing machine. The cartridge shall have no sharp edges and shall be designed to fit the embossing machine specified in the contract or order. (See 6.2.)

3.8.3 Reels. When furnished on reels, the tape shall be suitably wound so that the tape is easily and properly dispensed without binding or sticking when used in the embossing machine. The reel shall have no sharp edges and shall be designed to fit the embossing machine specified in the contract or order.

3.9 Workmanship. The finished tape shall be free from defects which detract from its appearance or may impair its servicability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For the purpose of sampling a lot shall consist of all tape of the same type, class, and color presented for inspection at one time as defined in MIL-STD-105.

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4.2.2 Sampling for examination of end item. Random samples shall be selected from each lot in accordance with MIL-STD-105. The lot size for purposes of determining the sample size shall be expressed in units of one roll, one cartridge, or one reel. No more than two rolls, randomly selected, shall be drawn from any one shipping container from each lot for each type, class, and color of tape offered for inspection for visual and dimensional characteristics. The same rolls of tape shall be used for examination under 4.3.1 and 4.3.2. The sample unit shall be one roll. The inspection levels and acceptable quality levels expressed in defects per 100 units shall be as follow:

Examination paragraph	Inspection level	AQL
4.3.1	S-4	2.5
4.3.2	S-1	6.5

4.2.3 Sampling for testing of end item. Random samples shall be selected from each lot in accordance with MIL-STD-105. The lot size for the purpose of determining the sample size shall be expressed in units of one roll, one cartridge, or one reel, as applicable. The inspection level shall be S-1, and the AQL 2.5 percent defective. The sample unit shall be two rolls, cartridges, or reels as applicable.

4.2.4 Sampling for preparation for delivery requirements. For examination of interior packages the sample unit shall be one shipping container fully prepared for delivery, and selected at random just prior to the closing operation. Closure defects shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection lot shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.3 Examination.

4.3.1 Samples taken in accordance with 4.2.2 shall be examined for defects listed in table III at AQL 2.5.

TABLE III. Examination of end item defects (AQL - 2.5)

Examine	Defect
Materials	
Backing	Not type and class specified.
Types I and II	Not coated on one side only with adhesive.
Cartridge (when applicable)	Missing Not designed to fit embossing machine as specified. Any sharp edges.
Reels (when applicable)	Not designed to fit embossing machine as specified. Any sharp edges.
Color (type I classes 2 and 3; types II and III, class 3)	Not color specified. Does not match standard.
Finish (type I)	Not finish specified.
General	Outside end of roll not secured by pressure-sensitive adhesive tape. Securing tape not easily removed; breaks or tears upon removal. Securing tape extends beyond edge of roll. Edges of roll sticky. Tape not wound liner side in.
Length	Not length specified.
Rolls	More than 2 inches in diameter.

4.3.2 Samples taken in accordance with 4.2.2 shall be examined for defects listed in table IV, at AQL 6.5.

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TABLE IV. Examination of end item defects (AQL - 6.5)

Examine	Defect
Liner	Missing, when specified. Not smooth. Does not completely cover adhesive. Liner breaks, tears or delaminates adhesive from backing upon removal.
Workmanship	Any tear, cut, hole, crack or crease. Embedded foreign matter. Edges ragged, nicked, crushed or uneven. Embossing tape binds or sticks while being unwound.
General	Tape not one continous piece. Any splice.

4.3.3 Inspection for preparation for delivery. Samples selected in accordance with 4.2.4 shall be examined for defects listed in table V, to determine that packaging, packing, and marking comply with the requirements of section 5. Defects of closure listed shall be examined on shipping containers fully prepared for delivery.

TABLE V. Classification of preparation for delivery defects

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged.
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.
Contents	Number of rolls or cartridges per container is less than required.
Weight	Weight of contents exceeds requirement.

4.4 Tests.

4.4.1 Conditioning. The tape shall be conditioned for testing in accordance with Method 4 of Federal Test Method Standard No. 147.

4.4.2 Test procedures.

4.4.2.1 Thickness. The thickness shall be determined as described in method 36 of Federal Test Method Standard No. 147.

4.4.2.2 Adhesion to steel, initial (types I and II only). The initial adhesion to steel shall be determined as described in method 2050 of Federal Test Method Standard No. 101. (See 3.5). Use a 15 minute dwell time between application of the tape to the test panel and the performance of the adhesion test.

4.4.2.3 Holding power (types I and II only). The holding power shall be determined in accordance with method 2054 of Federal Test Method Standard No. 101 with the following variations. (See 3.5).

Use a test specimen 1/2 inch in width and 6 inches long.
Use a 500 gram load.
Center the specimen, without added pressure on one end of the polished surface of the panel, to cover a 1/2 inch by 1 inch rectangle with tape.
Maintain the load for 4 hours.

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4.4.2.4 Embossing. Emboss the tape with the first seven letters of the alphabet with an embossing machine conforming to OO-E-491 and then visually examine from a distance of 3 feet to determine conformance with 3.5. In dispute cases, a machine with a new embossing wheel shall be used.

4.4.2.5 Transparency. A strip of the type I, class 1 tape shall be applied directly over black 10 point (1/8 inch) print on white paper and examined for legibility of the print (see 3.2).

4.4.2.6 Luminance. The type I, class 3 tape shall be tested for luminance according to MIL-L-3891. The luminance shall be determined when the tape is irradiated by the specified tungsten illumination (see 3.3.1).

4.4.2.7 Resistance to accelerated weathering. The tape shall be tested for colorfastness, legibility, and transparency, as applicable, after exposure for 100 hours in accordance with method 4050 of Fed. Test Method Std. No. 101 with the tape backing facing the light source.

4.4.2.7.1 Legibility. Prepare the specimen for legibility by embossing the first seven letters of the alphabet on the tape with an embossing machine conforming to OO-E-491. After exposure, examine for conformance with 3.5.

4.4.2.7.2 Transparency (type I Class 1). Hang specimen with liner independently in the apparatus so that ends do not curl. After exposure test the tape, with liner removed, as specified in 4.4.2.5. If two or more specimens break during the test, the tape shall be considered failing with respect to accelerated weathering. (see 3.5).

4.4.2.7.3 Colorfastness. Specimens prepared and exposed as specified in 4.4.2.7 shall be compared with an unexposed specimen and the colorfastness shall be evaluated as follows: (see 3.5).

Excellent - No perceptible change in color.
Good - Perceptible but no appreciable change in color.
Fair - Appreciable but no objectionable change in color.
Poor - Objectionable change in color.

Appreciable change in color means a change that is immediately noticeable in comparing the exposed specimen with the original comparison specimen. If closer inspection or a change of angle of light is required to make a slight change in color, the change is not appreciable. Objectionable change means a change that is more than appreciable to the extent that the departure from the original is so great that the color does not fall into the original color category, i.e., yellow to brown, red to pink.

4.4.2.8 Resistance to accelerated aging. Tape shall be tested for adhesive flow, colorfastness, liner removal, and adhesion after exposure in accordance with method 4047 of Fed. Test Method Std. No. 101, on a roll of tape placed in the desiccator in a horizontal position.

4.4.2.8.1 Adhesive flow. After exposure examine roll for excessive tackiness or oozing of the adhesive from the roll (see 3.5).

4.4.2.8.2 Adhesion. After conditioning in accordance with 4.4.2.8 test as specified in 4.4.2.2. (See 3.5).

4.4.2.8.3 Liner removal. Liner removal shall be determined by removing the liner from 6 inches of the tape and noting whether the liner is difficult to remove, breaks, tears, or delaminates any adhesive from backing. (See 3.5).

4.4.2.8.4 Colorfastness. After exposure evaluate for colorfastness as specified in 4.4.2.7.3. (See 3.5).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A (military requirements). Ten tapes of one description only shall be packaged in a snug-fitting folding paperboard box conforming to PPP-B-566, variety 1, style III, type edged, class 1; set-up paperboard box conforming to PPP-B-676., type I, variety 1, class A, style 4; or metal-paperboard box conforming to PPP-B-665, style A, class 1. Each box shall be closed in such a manner as to prevent accidental opening during shipment or handling.

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5.1.2 Level B (civil agencies). Ten tapes of one description only shall be packaged in a snug-fitting paperboard box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Each box shall be closed to prevent spillage of the contents.

5.1.3 Level C. Tape shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Tapes of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to PPP-B-636, grade V2s, style RSC. Each shipping container shall be closed, waterproofed and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B.

5.2.2.1 Military requirements. Tape in rolls or cartridges of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to PPP-B-636, type CF or SF, class domestic, variety SW, grade 275. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2.1.1 When specified, (see 6.2), the fiberboard shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

5.2.2.2 Civil agencies. Fifty unit packages of tape in rolls or cartridges of one description, packaged as specified in 5.1, shall be packed in a close-fitting fiberboard shipping container conforming to PPP-B-636, class domestic, grade 275. Each shipping container shall be closed in accordance with method II, as specified in the appendix of the container specification.

5.2.3 Level C. Tape, packed as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.3 Precautionary marking. The following precautionary marking applicable only for types I and II shall appear on two panels of each shipping container:

Recommended Storage Conditions

STORE BEHIND PRESENT STOCK

Store in a clean dry place. Temperature of 70 degrees Fahrenheit (21 degrees Centigrade) and 40 to 50 percent relative humidity are recommended.

ROTATE YOUR STOCK

6. NOTES

6.1 Intended use. The embossing tape covered by this specification is intended for use in embossing machines conforming to OO-E-491, for labeling, coding, identifying, and marking equipment in laboratories, warehouses, storage depots, offices and the like. The plastic tape is for general use in areas where it will not be exposed to temperatures under 0°F. or over 150°F. or to long periods of direct sunlight. The metal tape with adhesive may be used in the temperature range of 0°F. to 300°F. The metal tape without adhesives has only the temperature limitation of the particular metal being used.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and class of tape required (see 1.2).
(c) Color and finish required (see 3.3 and 3.4).
(d) Width and length required (see 3.6 and 3.7).
(e) Whether tape is to be furnished in cartridges on reels (see 3.8).
(f) Identification of machine in which cartridges will be used (see 3.8.2).
(g) Selection of applicable level of packaging and packing requires (see 5.1 and 5.2).
(h) Whether V-board shipping containers are required for level B shipments (see 5.2.2.1.1).

6.3 Type II, class 3 - Contrast aluminum, and Type III, class 3 - Contrast aluminum have been deleted.

MILITARY INTERESTS

Custodians:

Army - GL
Air Force - 84

Review activity:

Army - MD

User activities:

Army - EL, SM, MI
Navy - MC, EC

Preparing activity:

GSA-FSS

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